Benjamin Lobos Lertpunyaroj

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Education

Purdue University — **West Lafayette, IN**
Bachelor of Science in Computer Science — May 2027
Concentrations: Software Engineering, Systems Programming — GPA: 3.85/4.0
Minor: Business Management

- College of Science Deans List & Semester Honors: *2023, 2024, 2025*
- Relevant Coursework: *Data Structures and Algorithms (C++), OOP, Discrete Mathematics, Calculus, Linear Algebra, Software Engineering, Computer Architecture (Asm Arm64), Programming in C, Systems Programming, Operating Systems.*

Skills

Languages C, C++, C#.Net, Python, JavaScript (TS, React), HTML/CSS (TailwindCSS), Java, Swift, Lua, VimScript, R, Bash, AppleScript, Assembly (x86, Arm64)
Tools LaTeX, Vi/Vim/Neovim, Zsh, Unity, Airflow, Linux, Debian, PostgreSQL, AArch64, Ubuntu, Unix, Git

Experience

Embedded Software Engineer @ ESAP — **August 2025 - Present**

- Engineered a C++ Hardware-in-the-Loop (HIL) simulation engine on a Raspberry Pi, implementing Protobuf telemetry serialization to process simulated IMU, altimeter, and LiDAR data through a SPI connection for UAV controller testing.
- Architected a deterministic embedded data-exchange layer bridging the Raspberry Pi, ESP32-S3 (DUT), and Nucleo H563ZI using DMA-driven SPI and I2C protocols featuring CRC-validated framing and synchronized GPIO signaling.
- Led development on a modular Zephyr I2C target emulator for physical sensor spoofing, utilizing atomic primitives and snapshot caching.

C Programming Teaching Assistant @ Purdue University — **November 2024 – Present**

- Provided academic support to 750+ students (CS 24000), leading lab sessions, holding weekly office hours, and clarifying concepts related to the C programming language
- Developed homework assignments with custom built test-cases in C to reinforce data structure fundamentals, writing up handouts using Latex.
- Implemented multiple midterm/final practice exams into the class curriculum to enhance students' exam readiness and practical skills.

Software Engineering Intern @ Qservus — **December 2022 – March 2023**

- Shipped production-facing frontend features for Qservus Lite, a customer feedback and service-quality platform, using AngularJS.
- Enhanced deployment workflows for the international launch of Qservus's main web platform, improving release efficiency and readiness for broader adoption.
- Developed a native iOS survey application in Swift, expanding mobile product capabilities and streamlining customer feedback collection.

Projects

BoilerMate — TypeScript (React), PostgreSQL, Supabase, Prisma ORM, NestJS, AWS, AI — **August 2025 – January 2026**
Roommate Matching Solution

- Designed a roommate discovery website to streamline Purdue University's off-campus housing search, leveraging a NestJS/PostgreSQL backend to deliver a social media implementation powered by LLMs (Google AI Studio).
- Led coordination and onboarding of team members through Agile/SCRUM principles, shipping with over 5,000+ impressions and partnering with Purdue faculty stakeholders for expanding outreach.

Elite Edge — JavaScript (React), TailwindCSS, Python, PostgreSQL, Apache Airflow, Nginx — **May 2025 - February 2026**
College Basketball Weekly Insights & Analytics

- Architected a Dockerized backend to scrape weekly KenPom data (2013–2025), store it in PostgreSQL, and schedule a Python (scikit-learn) model running with Apache Airflow and a custom backend Nginx server for delivering NCAA advanced analytics.
- Constructed Machine Learning models (KNN, XGBoost) analyzing 12 years of data with 30+ features to predict tournament team selection and Final Four advancement.

Custom UNIX Shell — C++, C, UNIX, Flex, Bison — **September – November 2025**
Shell Command Line Tool for UNIX-based OS

- Built a Unix-based shell in C++ that tokenizes/parses user input (Flex/Bison) and executes commands via POSIX system calls, supporting arguments and standard I/O redirection.
- Engineered support for advanced features including interrupts, zombie process handling, builtin functions, environment variables, regex, and history, with an optimized memory allocator for running processes made in C.